Introduction to Actuarial Audit under IFRS 17

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Agenda

1. Prolog: What is Actuarial Audit ?
2. Actuarial Science and Accounting
3. Overview of IFRS 17
4. Recognition and Measurement under IFRS 17
    - Level of Aggregation
    - Discount Rate
    - Risk Adjustment
    - Contractual Service Margin (CSM)
    - Measurement Model
    - Transition Approach
5. Remeasurement
    - Liability for Incurred Claim (LfIC)
6. Financial Disclosure and Statement Mapping

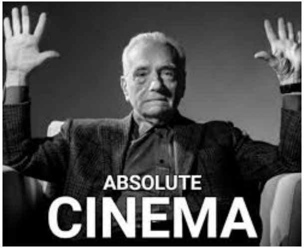
My Personal Goal from Today's Session I

1. Sharing 30% accounting knowledge 70% actuarial application
2. Showing you some memes along the presentation so keep your eyes open :D
3. Audit is about how you think, not just what you do.
4. Prevent one of these things to happen to you during your first internship/job

(Thanks to: John Lee)

My Personal Goal from Today's Session II

5. Your natural reaction after the sharing session:



Out of Scope

1. Making you an IFRS17 wizard
2. Time Value of Options and Guarantees (TVOG)
3. Reinsurance Contract Held
4. Shariah Insurance
5. Goodwill
6. Insurance Merger and Acquisition (M&A)

Disclaimer

Please read

- **Purpose:** knowledge sharing / general educational content.
- **Not advice:** this is *not* actuarial, accounting, audit, or legal advice.
- **Use at your own risk:** any reliance or decisions are the user's responsibility.
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Prolog: What is Actuarial Audit ?

IFRS 17 actuarial audit

Audit and Actuarial Audit

An audit in finance and accounting is a comprehensive examination of an organization's financial records conducted by qualified professionals. These experts meticulously review financial statements to confirm their accuracy, ensure compliance with applicable regulations, and corroborate that the information fairly represents the organization's **financial position**. This includes the **income statement**, **balance sheet**, and cash flow statement.[1]
Actuarial audit is just the subset of audit where the object audited is mostly provided or done by actuaries.

[1] Investopedia

Example of IFRS17 Financial Statement - Balance Sheet - Life Insurance

Below is the Prudential Life snapshots of IFRS17 disclosure access on 6 February 2026.

	Note	2025 $m 30 Jun	2024 $m 30 Jun	2024 $m 31 Dec
Assets				
Goodwill	C4.1	**889**	819	848
Other intangible assets	C4.2	**3,939**	3,758	3,824
Property, plant and equipment	C1.2	**537**	390	417
Insurance contract assets	C3.1	**1,722**	1,131	1,345
Reinsurance contract assets	C3.1	**3,267**	3,200	3,390
Deferred tax assets		**150**	155	142
Current tax recoverable		**75**	25	31
Investments in joint ventures and associates accounted for using the equity method		**2,517**	1,781	2,412
Investment properties	C1.1	**3**	3	3
Loans	C1.1	**534**	543	517
Equity securities and holdings in collective investment schemes [note]	C1.1	**83,705**	73,110	81,002
Debt securities [note]	C1.1	**84,871**	74,543	73,804
Derivative assets	C1.1	**1,528**	276	395
Deposits	C1.1	**6,141**	5,284	5,466
Accrued investment income	C1.2	**1,039**	960	902
Other debtors	C1.2	**2,284**	2,440	1,310
Assets held for sale	C1.2	**282**	291	296
Cash and cash equivalents	C1.1	**5,636**	5,978	5,772
Total assets		**199,119**	174,687	181,876

	Note	2025 $m 30 Jun	2024 $m 30 Jun	2024 $m 31 Dec
Equity				
Shareholders' equity	C3.1	**18,119**	16,171	17,492
Non-controlling interests		**1,325**	1,065	1,182
Total equity		**19,444**	17,236	18,674
Liabilities				
Insurance contract liabilities	C3.1	**161,476**	141,099	147,566
Reinsurance contract liabilities	C3.1	**510**	1,379	536
Investment contract liabilities without discretionary participation features	C2.2	**730**	819	748
Core structural borrowings of shareholder-financed businesses	C5.1	**4,473**	3,930	3,925
Operational borrowings	C5.2	**853**	961	797
Obligations under funding, securities lending and sale and repurchase agreements		**665**	576	272
Net asset value attributable to unit holders of consolidated investment funds		**2,332**	2,921	2,679
Deferred tax liabilities		**1,772**	1,339	1,514
Current tax liabilities		**273**	231	238
Accruals, deferred income and other creditors	C1.2	**5,235**	3,395	2,848
Provisions		**157**	137	218
Derivative liabilities		**924**	426	1,617
Liabilities held for sale	C1.2	**275**	238	244
Total liabilities		**179,675**	157,451	163,202
Total equity and liabilities		**199,119**	174,687	181,876

Note
Included within equity securities and holdings in collective investment schemes and debt securities as at 30 June 2025 are $1,599 million of lent securities and assets subject to repurchase agreements (30 June 2024: $1,680 million; 31 December 2024: $1,565 million).

Observe that the Insurance Contract Liabilities (ICL) are around 80% of the total equity and liabilities, which makes it more important to verify whether the amount is correct.

Example of IFRS17 Financial Statement - Profit and Loss - Life Insurance

	Note	2025 $m Half year	2024 $m Half year	2024 $m Full year
Insurance revenue	B1.4	**5,326**	4,961	10,358
Insurance service expenses		**(3,961)**	(3,638)	(7,763)
Net expense from reinsurance contracts held		**(125)**	(252)	(302)
Insurance service result		**1,240**	1,071	2,293
Investment return	B1.4	**7,059**	2,495	5,919
Fair value movements on investment contract liabilities		**(14)**	(54)	(95)
Net insurance and reinsurance finance expense		**(6,149)**	(2,274)	(4,492)
Net investment result		**896**	167	1,332
Other revenue	B1.4	**189**	197	382
Non-insurance expenditure		**(487)**	(532)	(1,003)
Finance costs: interest on core structural borrowings of shareholder-financed businesses		**(87)**	(85)	(171)
Loss attaching to corporate transactions	B1.1	**(16)**	(69)	(71)
Share of (loss) profit from joint ventures and associates, net of related tax		**(28)**	(243)	477
Profit before tax *(being tax attributable to shareholders' and policyholders' returns)* note		**1,707**	506	3,239
Tax charge attributable to policyholders' returns		**(7)**	(112)	(286)
Profit before tax attributable to shareholders' returns	B1.1	**1,700**	394	2,953
Total tax charge attributable to shareholders' and policyholders' returns	B2	**(348)**	(324)	(824)
Remove tax charge attributable to policyholders' returns		**7**	112	286
Tax charge attributable to shareholders' returns		**(341)**	(212)	(538)
Profit for the period		**1,359**	182	2,415

Top 4 of the largest line items are investment return, net insurance and reinsurance finance expense, insurance revenue and insurance service expenses

Example of IFRS17 Financial Statement - Balance Sheet - General Insurance

Below is the Zurich Insurance snapshots of IFRS17 disclosure access on 9 February 2026.

Consolidated balance sheets

Assets

in USD millions, as of December 31	Notes	2024	2023
Assets			
Cash and cash equivalents		**6,768**	**7,280**
Total Group investments	5	152,562	140,966
Equity securities		14,182	13,217
Debt securities		118,415	105,924
Investment property		11,734	13,684
Mortgage loans		4,047	4,324
Other financial assets		4,039	3,682
Investments in associates and joint ventures		146	135
Investments for unit-linked contracts		148,535	141,144
Total investments		**301,098**	**282,110**
Insurance contract assets	7	768	580
Reinsurance contract assets	7	21,450	21,942
Receivables and other assets	14	11,717	10,391
Deferred tax assets	16	1,703	1,700
Assets held for sale[1]	4	1,203	23,758
Property and equipment	12	1,867	2,092
Attorney-in-fact contracts	13	2,650	2,650
Goodwill	13	4,805	4,541
Other intangible assets	13	3,977	4,337
Total assets		**358,005**	**361,382**

1 As of December 31, 2024, the Group had USD 1.2 billion of assets held for sale based on agreements signed to sell portfolios of Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4). In 2023, the Group had USD 23.8 billion of assets held for sale based on agreements signed to sell portfolios of Zurich Deutscher Herold LebensversicherungAktiengesellschaft, Zurich Chile Seguros de Vida S.A., Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4).

Liabilities and equity

in USD millions, as of December 31	Notes	2024	2023
Liabilities			
Liabilities for investment contracts	8	66,507	60,270
Insurance contract liabilities	7	230,479	216,962
Reinsurance contract liabilities	7	437	504
Obligation to repurchase securities		1,123	796
Other liabilities[1]	15	16,022	16,661
Deferred tax liabilities	16	2,446	2,300
Liabilities held for sale[2]	4	1,162	23,860
Senior debt	17	4,020	5,190
Subordinated debt	17	8,871	8,559
Total liabilities		**331,067**	**335,102**
Equity			
Share capital	18	10	10
Additional paid-in capital	18	1,410	1,333
Net unreal. gains/(losses) on financial assets		(4,452)	(4,307)
Change in discount rate for (re)insurance contract		4,372	4,291
Change in fair value of underlying items		1,158	1,053
Cumulative foreign currency translation adjustment		(11,103)	(10,616)
Revaluation reserves		254	254
Retained earnings		33,823	32,842
Shareholders' equity		25,472	24,860
Non-controlling interests		1,466	1,419
Total equity		**26,938**	**26,280**
Total liabilities and equity		**358,005**	**361,382**

1 Includes restructuring provisions, litigation and regulatory provisions (see note 15) and other provisions.
2 As of December 31, 2024, the Group had USD 1.2 billion of liabilities held for sale based on agreements signed sell portfolios of Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4). In 2023, the Group had USD 23.9 billion of liabilities held for sale based on agreements signed to sell portfolios of Zurich Deutscher Herold LebensversicherungAktiengesellschaft, Zurich Chile Seguros de Vida S.A.,Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4).

Observe that the Insurance Contract Liabilities (ICL) is around 60% of the total equity and liabilities, which make it more important to verify whether the amount is correct.

Example of IFRS17 Financial Statement - Profit and Loss - General Insurance

Consolidated income statements

in USD millions, for the years ended December 31	Notes	2024	2023
Insurance revenue	9	59,507	56,099
Insurance service expense		(50,456)	(47,422)
Net expenses from reinsurance contracts held		(3,031)	(2,981)
Insurance service result		6,020	5,696
Net investment income on Group investments		5,730	5,382
Net capital gains/(losses) on Group investments		1,084	(696)
Net investment result on Group investments	5	6,814	4,687
Net investment result on unit-linked investments		16,384	14,191
Change in liabilities for investment contracts and other funds		(8,112)	(6,378)
Re-/insurance finance income/(expenses)		(12,244)	(10,963)
Net investment result		2,842	1,536
Fee income	10	6,011	5,885
Fee business expenses	10	(3,575)	(3,583)
Fee result		2,436	2,303
Other revenues		358	210
Net gains/(losses) on divestment of businesses	4	114	(104)
Interest expense on debt		(440)	(456)
Other expenses	11	(2,864)	(2,727)
Other result		(2,832)	(3,077)
Net income before income taxes		8,465	6,458
of which: Attributable to non-controlling interests		566	536
Income tax (expense)/benefit	16	(2,261)	(1,741)
attributable to policyholders		(179)	(172)
attributable to shareholders		(2,082)	(1,568)
of which: Attributable to non-controlling interests		(176)	(170)
Net income after taxes		**6,204**	**4,717**
attributable to non-controlling interests		390	366
attributable to shareholders		5,814	4,351

Audit Challenge I

Insurance company usually relies on software that deals with hundreds of thousands or even billions of rows of data and there is no way one can check all the data by the naked eye. This gives another challenge to auditors where the engagement is **relatively short** (3–5 months), but needs to verify all the business activities done by the company for a **certain period** (usually quarterly, semi-annually or annually). This means an auditor needs to be efficient with time, focus on high-risk items and communicate clearly what they need from the company.
There are 3 approaches that auditor can do to verify the financial statement, i.e., **population**, **sample** and **analytical** and the most important item require before the audit is **data reconciliation**.
Due to the size of data, auditor need to first verify the data given by the company. Whether it is

1. **consistent**, does the amount noted is the same with the previous report or other document ?

Audit Challenge II

2. **sensible**, e.g., does it normal for a life insurance to have a high amount of claim reserve ? does the change is similar with last year ?
3. **accurate**, does the amount in the financial statement is correctly copied or correctly calculated?
4. **complete**, has all the component in certain accounting item has fully calculated?

We can agree that it is tedious work to reconcile data over and over again. A particular tool that can help to capture/record the movement of the data is by doing a **waterfall chart** or a *walk* from the initial or verified amount into the most recent amount. A simple walk of number of policies can be seen below:

Audit Challenge III

Table: Policy Count Movement — Full Year 202X

Item	Movement	Policies in Force
Opening Balance (1 Jan 202X)		100,000
New Business	+12,000	112,000
Deaths	–500	111,500
Surrenders	–3,000	108,500
Maturities	–4,500	104,000
Closing Balance (31 Dec 202X)	+4,000	104,000

My Actuarial Audit Experience I

In my experience, the audit project usually goes like this

1. Kick-off meeting: where we align the scope of audit, the procedure to verify the numbers, project timeline and last year finding/observation (if any)
2. Initial data request: product listing of the Insurance Contract Liabilities (ICL), trial balance, assumptions and model changes documentation.
3. Reconciliation between the product listing and the trial balance, and the walk or changes happened from last audit/year to current year.
4. Assumption checks: where we review the assumptions used for the current year calculation. The major assumptions are: mortality rate, morbidity rate, lapse/surrender rate, expense assumptions, and discount rate.
5. New Business assessment: profit tagging, measurement model used, and portfolio/product grouping.

My Actuarial Audit Experience II

6. Sampling process: where we pick samples to check. The samples are consist of, (Re)Insurance Contract Groups (ICG/RCG), product, and policies.
7. Model Baselining: where we recalculate or assess the reasonableness of the reserve calculation in policy and ICG level. Insurance companies usually use, Prophet, RAFM (Risk Agile FM), and AXIS.
8. Financial Disclosure: where we reconcile the changes due to remeasurement is correctly presented in the financial disclosure

Audit is A State of Mind

Sure these things are the responsible for an auditor to check and the auditor will get paid by auditing the full year data. But who really responsible to ensure the data or the calculation done correctly?

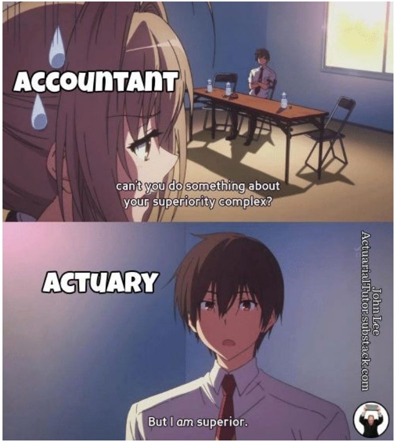
The company!

We can agree that all these calculation or this reporting is done by human who gives input to a machine and therefore it will always expose to human error. But by having a view on auditor as a co-worker rather than a piggybacking person, it can significantly improves the productivity of the company.
By having a well documented workflow, auditors can easily verify the result calculated and the actuaries can do more productive work.

Actuarial Science and Accounting

IFRS 17 actuarial audit

Actuaries and Accountants



Why do these memes exist? Is it true? (Credit to:John Lee)

Accounting Standards for Insurance

In (re)insurance one of the International Financial Reporting Standards (IFRS) we use is IFRS 17 : Insurance Contract. In every IFRS, it has 3 major sections:

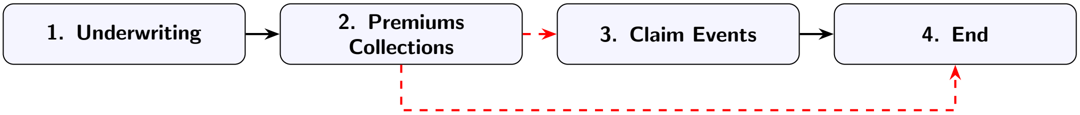
1. Scope: which items are covered by the IFRS (accounting),
2. Recognition and Measurement: how to record the initial balance (actuarial science),
3. Disclosure: how to present the result in financial disclosures (accounting).

Doing the actuarial audit, means that we need to check the **documentation, assumptions used** and whether it align with the **standards**.
It is said that doing IFRS17 would required at least 3 parties, i.e., accounting, actuarial, and IT. We are going to see in this session why this will be the case.

Overview of IFRS 17

IFRS 17 actuarial audit

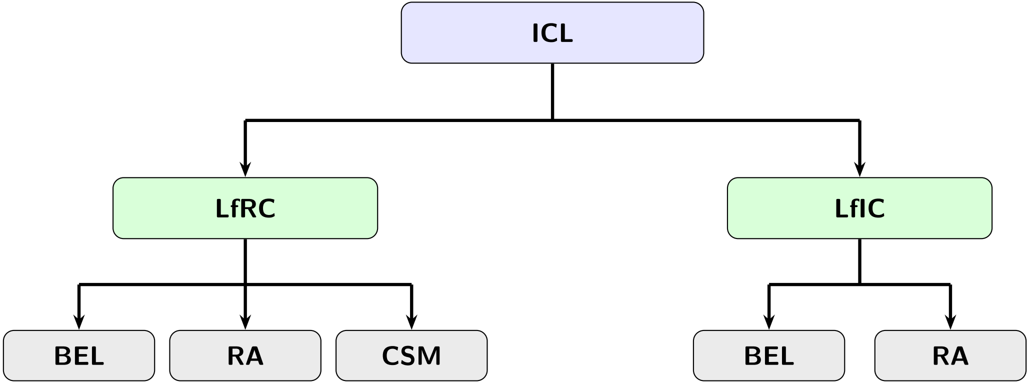
Life of a Premium



We are going to take a closer look on the **lifecyle** of an insurance from previous slide and ask ourselves, "what happened to the money after policyholders pay the premium? Does the insurer just keep the money in a bank? Pay claims to other policyholder? or just hope that no claim will submitted ?

When the insurance is **issued**, the contract generally becomes a **liability** to the insurer meaning that the insurance company is liable to pay the policyholder if a claim event happens. But it doesn't mean that every contract will be a liability to the insurance company; there are cases where the contract is presented as an asset (as in the previous balance sheet examples).

What's inside the Insurance Contract Liability (ICL)?



Some Quick Definitions I

Liability for Remaining Coverage (LfRC): is an entity's obligation to:

1. investigate and pay valid claims under existing insurance contracts for insured events that **have not yet occurred** (i.e., the obligation that relates to the unexpired portion of the insurance coverage); and
2. pay amounts under existing insurance contracts that are not included in (1) and that relate to:
    1. insurance contract services not yet provided (i.e., the obligations that relate to future provision of insurance contract services); or
    2. any investment components or other amounts that are not related to the provision of insurance contract services and that have not been transferred to the Liability for Incurred Claims (LfIC)

Source: IFRS 17, Appendix A (Definitions)

Some Quick Definitions II

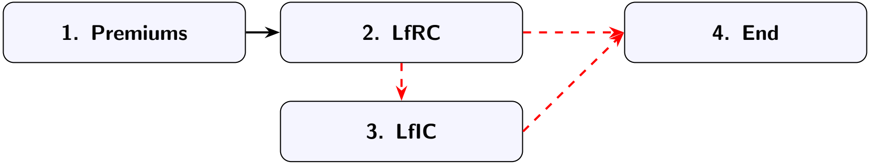
Liability for Incurred Claims (LfIC): is an entity's obligation to:

1. investigate and pay valid claims for insured events that have **already occurred**, including events that have occurred but for which claims have not been reported, and other incurred insurance expenses; and
2. pay amounts that are not included in (1) and that relate to:
   1. insurance contract services that have already been provided; or
   2. any investment components or other amounts that are not related to the provision of insurance contract services and that are not in the Liability for Remaining Coverage (LfRC).

Source: IFRS 17, Appendix A (Definitions)

What happened to the premium so far

Based on the explanation so far, we can see that the paid premium will directly become an LfRC and the LfRC will be an LfIC if claim event happened.



In the next slides, we will go deeper into how to calculate LfRC and LfIC.

Recognition and Measurement under IFRS 17

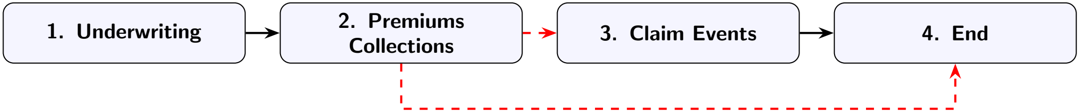
IFRS 17 actuarial audit

Introduction to Insurance Contract

Insurance contract is a contract under which one party (the issuer) accepts **significant insurance risk** from another party (the policyholder) by agreeing to **compensate** the policyholder if a specified **uncertain future event** (the insured event) adversely affects the policyholder.

Source: IFRS 17, Appendix A (Definitions)

Typical lifecycle of an insurance as a policyholder:



But how to determine the **significant insurance risk**?

What is an insurance contract (conceptually)? I

Insurance risk is significant if and only if, an insured event could cause the issuer to pay **additional amounts** that are significant in any single scenario, excluding scenarios that have no commercial substance (i.e., no discernible effect on the economics of the transaction) ...

Source: IFRS 17, Appendix B18

The additional amounts above are determined on a present-value basis.

Source: IFRS 17, Appendix B20

What is an insurance contract (conceptually)? II

The additional amounts refer to the present value of amounts that exceed those that would be payable if no insured event had occurred (excluding scenarios that lack commercial substance). Those additional amounts include claims handling and assessment costs, but exclude:

- the loss of the ability to charge the policyholder for future service,
- a waiver, on death, of charges that would be made on cancellation or surrender,
- a payment conditional on an event that does not cause a significant loss to the holder of the contract,
- possible reinsurance recoveries.

Source: IFRS 17, Appendix B21

Or as an actuary, this means to:

$$\frac{\sum_i E\left[\text{PVCF}_{i,\text{ claim event}}\right]}{\sum_i E\left[\text{PVCF}_{i,\text{ no claim event}}\right]} \tag{1}$$

Quick Break

From this section, now you know how to answer this question from a very famous *"star"*

Level of Aggregation (LoA)

In order to calculate the LfRC and LfIC, IFRS17 requires to aggregate (group) insurance contracts into portfolios. IFRS 17 is clear on how contracts should be grouped. There are three criteria:

- Similar risk: product type or characteristics (e.g., premium payment mode),
- Profitability: expected profitability of the product,
- Annual cohort: issue year of the insurance contract. IFRS17 prohibits to include contracts issued more than one year apart in the same group

Source: IFRS 17, Para. 14, 16, 22

This aggregation is done only at the initial recognition of the contract.
Source: IFRS 17, Para. 24

Level of Aggregation (LoA) - Profitability I

IFRS17 requires that the product is divided at least into 3 profitability group i.e.,:

- Onerous at initial recognition
- No Significant Possibility of Becoming Onerous (NSPBO)
- Remaining

Source: IFRS 17, Para. 16

IFRS17 defines onerous insurance contract at the date of initial recognition if the **Fulfilment Cash Flows (FCF)** allocated to the contract, any previously recognised **Insurance Acquisition Cash Flows (IACF)** and any cash flows arising from the contract at the date of initial recognition in total are a **net outflow**.

Source: IFRS 17, Para. 47

Level of Aggregation (LoA) - Profitability II

There is no specific guideline from IFRS 17 in determining NSPBO and remaining category, but it requires an insurance company to assess the methodology to tag NSPBO or remaining category based on

- likelihood of changes in assumptions which, if they occurred, would result in the contracts becoming onerous
- using information about estimates provided by the entity's internal reporting

Source: IFRS 17, Para. 19

What to Measure

On initial recognition, an entity shall measure a group of insurance contracts at the total of:

- the Fulfilment Cash Flows (FCF), which comprise:
  - estimates of future cash flows
  - an adjustment to reflect the time value of money and the financial risks related to the future cash flows, to the extent that the financial risks are not included in the estimates of the future cash flows, and
  - Risk Adjustment (RA) for non-financial risk
- the Contractual Service Margin (CSM)

Source: IFRS 17, Para. 32

We can re-write FCF in a more familiar term to be:

$$\text{FCF} := E\left[\text{PVCF}\right] + \text{RA} = \text{BEL} + \text{RA} \tag{2}$$

where BEL stands for **Best Estimate Liability**. However we are going to discuss on, which future cash flow should be included in BEL calculation, how to choose the **discount rate** and how to calculate **RA**.

Cashflow in Scope Boundary I

Cash flows within the boundary of an insurance contract are those that relate directly to the fulfilment of the contract, including cash flows for which the entity has discretion over the amount or timing. IFRS 17 provides the following examples of such cash flows:

1. Premiums
2. Payments, including claims, to a policyholder
3. Payments to a policyholder that vary based on underlying items
4. Payments to a policyholder resulting from derivatives
5. Insurance acquisition cash flows
6. Claims handling costs
7. Costs incurred in providing contractual benefits in kind
8. Policy administration and maintenance costs
9. Transaction-based taxes and levies

Cashflow in Scope Boundary II

10. Payments by the insurer of tax in a fiduciary capacity
11. Potential cash inflows from recoveries
12. An allocation of fixed and variable overheads
13. Costs the entity will incur in providing an investment activity, an investment-return service or an investment-related service
14. Any other costs specifically chargeable to the policyholder

Source: IFRS 17, Para. B65

Audit Checkbox

We usually check the assumptions are consistent with the experience study and all the calculated cashflow has aligned with the product specification or not.

Discount Rate

The discount rates applied to the estimates of the future cash flows described previously shall:

1. reflect the time value of money, the characteristics of the cash flows and the **liquidity characteristics** of the insurance contracts;
2. be consistent with observable current market prices (if any) for financial instruments with cash flows whose characteristics are consistent with those of the insurance contracts, in terms of, for example, timing, currency and liquidity; and
3. exclude the effect of factors that influence such observable market prices but do not affect the future cash flows of the insurance contracts.

Source: IFRS 17, Para. 36

Discount Rate should be simple, isn't it ? I

Well yes, but according to IFRS 17, an entity shall use the a specific discount rate to measure or determine specific item. For example:

1. to measure the FCF: **current** discount rates
2. to determine the interest to accrete on the CSM for insurance contracts without direct participation features: discount rates determined at the **date of initial recognition** of a group of contracts.
3. to measure the changes to the CSM for insurance contracts without direct participation features of a group of insurance contracts for changes in FCF that relate to future,
    1. experience experience adjustments arising from premiums received in the period that relate to future service, and related cash flows such as insurance acquisition cash flows and premium-based taxes
    2. changes in estimates of the present value of the future cash flows in the LfRC except the effect of the time value of money and changes in the time value of money and the effect of financial risk and changes in financial risk
    3. changes in the RA for non-financial risk that relate to future service

Discount Rate should be simple, isn't it ? II

shall use discount rates determined on **initial recognition**

4. for groups of contracts applying the premium allocation approach that have a significant financing component, to adjust the carrying amount of the LfRC applying: discount rates determined on **initial recognition**
5. if an entity chooses to disaggregate Insurance Finance Income or Expenses (IFIE) between profit or loss and Other Comprehensive Income (OCI), to determine the amount of the IFIE included in profit or loss:
    1. for groups of insurance contracts for which changes in assumptions that relate to financial risk **do not have a substantial effect** on the amounts paid to policyholders: discount rates determined at the **date of initial recognition** of a group of contracts, to nominal cash flows that do not vary based on the returns on any underlying items;
    2. for groups of insurance contracts for which changes in assumptions that relate to financial risk **have a substantial effect** on the amounts paid to policyholders, applying paragraph B132(a)(i)—discount rates that allocate the remaining revised expected finance income or expenses over the remaining duration of the group of contracts at a **constant rate**; and

Discount Rate should be simple, isn't it ? III

3. for groups of contracts applying the premium allocation approach: discount rates determined at the date of the **incurred claim**

Source: IFRS 17, Para. B72

So in fact, we will use different kind of discount for a different calculation.

If you remember our previous slide, we learned that insurance contract can be grouped into one as long as the contracts issued no more than one year apart in the same group, so how do we determined the discount rate since there are several cases where the discount rate used is the discount rate determined at the date of initial recognition ? According to IFRS 17, para B73, an entity may use weighted-average discount rates over the period that contracts in the group are issued.

It is also mentioned that some calculations are required to be calculated using discount rate which determined at the date of initial recognition, how to determine this discount rate?

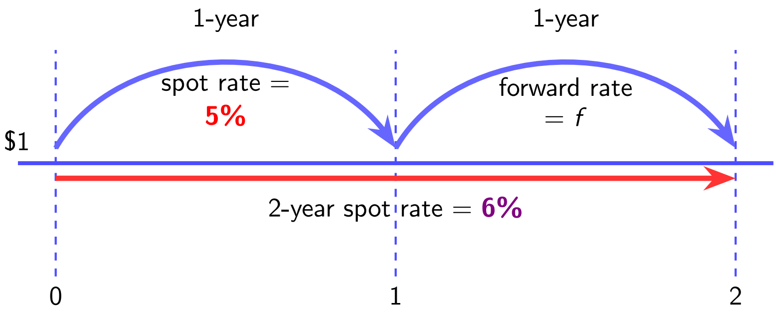
Spot Rate and Forward Rate I

We define spot rate s_t to be the annual effective interest rate earned by money invested **now** (time 0) for a period of t years. (Spot rates are sometimes called zero-coupon rates.) While (implied) forward rates f_t is the annual effective interest rate earned by money invested t_1 for a period of t_2 years. Mathematically we can formulate this to:

$$1 + f_{[t_1,t_2]} = \frac{(1 + s_{t_2})^{t_2}}{(1 + s_{t_1})^{t_1}} \tag{3}$$

Simple illustration on the calculation

Spot Rate and Forward Rate II



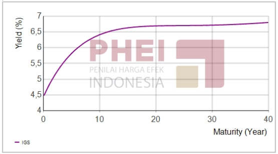
Spot Rate and Forward Rate III

In Indonesia, the yield rate data is published by Indonesia Bond Pricing Agency (IBPA). Due note that the rate in this website is **not** a zero-coupon rates, therefore we need to calculate the spot rate by ourselves using bootstraping method.

IGSYC | Benchmark and Retail Fair Prices | Corporate Bond Yield by Tenor

Indonesia Government Securities Yield Curve

per 19-Februari-2026

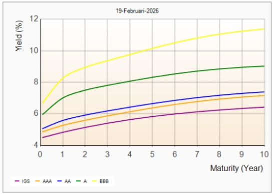


Tenor Year	Today	Yesterday	Tenor Year	Today	Yesterday
0,1	4,4931	4,5078	16,0	6,6544	6,6411
1,0	4,8239	4,8234	17,0	6,6694	6,6576
2,0	5,1358	5,1232	18,0	6,6804	6,6699
3,0	5,4031	5,3822	19,0	6,6884	6,6790
4,0	5,6312	5,6051	20,0	6,6939	6,6855
5,0	5,8251	5,7959	21,0	6,6978	6,6901
6,0	5,9890	5,9586	22,0	6,7003	6,6934
7,0	6,1268	6,0965	23,0	6,7021	6,6958
8,0	6,2421	6,2128	24,0	6,7035	6,6978
9,0	6,3378	6,3101	25,0	6,7048	6,6997
10,0	6,4167	6,3911	26,0	6,7062	6,7018
11,0	6,4813	6,4578	27,0	6,7080	6,7043
12,0	6,5335	6,5123	28,0	6,7103	6,7076
13,0	6,5754	6,5564	29,0	6,7133	6,7116
14,0	6,6086	6,5917	30,0	6,7171	6,7166
15,0	6,6345	6,6195	31,0	6,7217	6,7227

IGSYC | Benchmark and Retail Fair Prices | Corporate Bond Yield by Tenor

Corporate Bond Yield by Tenor

Tenor (Year)	IGS	AAA	AA	A	BBB
0,1	4,4931	4,8600	5,0601	5,9550	6,6911
1,0	4,8239	5,2647	5,5757	7,0036	8,2612
2,0	5,1358	5,5882	5,9180	7,4976	8,9502
3,0	5,4031	5,8693	6,1833	7,8049	9,3728
4,0	5,6312	6,1319	6,4245	8,0724	9,7666
5,0	5,8251	6,3749	6,6491	8,3185	10,1545
6,0	5,9890	6,5920	6,8523	8,5339	10,5115
7,0	6,1268	6,7790	7,0286	8,7107	10,8164
8,0	6,2421	6,9346	7,1761	8,8484	11,0614
9,0	6,3378	7,0606	7,2954	8,9510	11,2492
10,0	6,4167	7,1602	7,3893	9,0248	11,3880



Bottom Up and Top Down

IFRS 17 proposes two basic methods for determining discount rates for cash flows of insurance contracts that do not vary based on the returns on underlying items, i.e., bottom-up approach and top-down approach.
In practice the top-down approach and the bottom-up approach may result in different yield curves, even in the same currency
Source: IFRS 17, Para. B84

Audit Checkbox

We usually verify whether the rates determination and used is already correct and consistent throughout document.

Bottom-Up Approach

Bottom-up approach: determines discount rates by adjusting a **liquid risk free** yield curve to reflect the differences between the liquidity characteristics of the **financial instruments that underlie the rates observed** in the market and the liquidity characteristics of the insurance contracts

Source: IFRS 17, Appendix B80

Top-down Approach

Top-down approach: determines the appropriate discount rates for insurance contracts based on a yield curve that reflects the **current market** rates of return implicit in a fair value measurement of a **reference portfolio** of assets. An entity should adjust that yield curve to eliminate any factors that are not relevant to the insurance contracts, but is not required to adjust the yield curve for differences in liquidity characteristics of the insurance contracts and the reference portfolio.
Source: IFRS 17, Para. B81

Risk Adjustment I

Risk Adjusment for non-financial risk (RA) is the compensation an entity requires for bearing the uncertainty about the amount and timing of the cash flows that arises from non-financial risk as the entity fulfils insurance contracts. *Source: IFRS 17, Appendix A (Definitions).*

The risk adjustment for non-financial risk for insurance contracts measures the compensation that the entity would require to make the entity indifferent between:

1. fulfilling a liability that has a **range of possible outcomes** arising from non-financial risk; and
2. fulfilling a liability that will **generate fixed cash flows** with the same expected present value as the insurance contracts.

Source: IFRS 17, Appendix B87

Because the risk adjustment for non-financial risk reflects the compensation the entity would require for bearing the non-financial risk arising from the uncertain amount and timing of the cash flows, the risk adjustment for non-financial risk also reflects:

Risk Adjustment II

1. the degree of **diversification benefit** the entity includes when determining the compensation it requires for bearing that risk; and
2. both favourable and unfavourable outcomes, in a way that reflects the entity's **degree of risk aversion**.

Source: IFRS 17, Appendix B88

IFRS 17 does not specify the estimation technique(s) used to determine the RA. However, to reflect the compensation the entity would require for bearing the non-financial risk, the RA shall have the following characteristics:

1. risks with low frequency and high severity will result in higher risk adjustments for non-financial risk than risks with high frequency and low severity;
2. for similar risks, contracts with a **longer duration** will result in **higher RA** than contracts with a shorter duration;

Risk Adjustment III

3. risks with a **wider probability distribution** will result in **higher RA** than risks with a narrower distribution;
4. the **less that is known** about the current estimate and its trend, **the higher will be the RA**; and
5. to the extent that **emerging experience reduces uncertainty** about the amount and timing of cash flows, **RA will decrease** and vice versa.

Source: IFRS 17, Appendix B91

Most common techniques used in calculating RA is confidence level, Value at Risk (VaR), and Cost of Capital (CoC).

Audit Checkbox

We usually check the RA methodology is correct statistically and calculation is correct.

Contractual Service Margin (CSM) and Loss Component (LC) I

This is the bread and butter, the totally new concept introduced in the IFRS17. Contractual Service Margin (CSM) is a component of the carrying amount of the asset or liability for a **group of insurance contracts** representing the **unearned profit** the entity will recognise as it provides **insurance contract services** under the insurance contracts in the group.
Source: IFRS 17, Appendix A (Definitions)

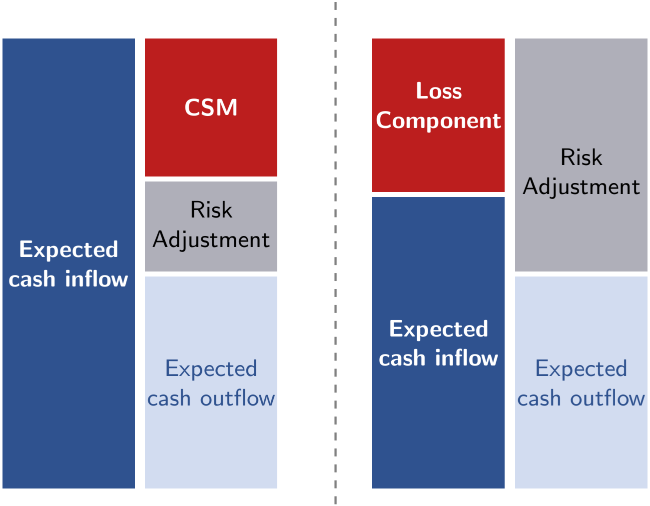
At initial recognition, if the group of contract is not onerous, CSM is calculated:

$$\text{CSM} := \text{PVCF}_{\text{inflow}} - \text{PVCF}_{\text{outflow}} - \text{RA} \tag{4}$$

If we look closely, this formula looks like the FCF calculation that we saw earlier. This is definitely intentional where it implicate that there is no profit to be recognized at inception. We also saw in earlier section, we learned that even at initial recognition there might be insurance contract where it is not profitable. This only happened if the CSM is negative and it will be called as Loss Component (LC). We will revisit this in detail later in the remeasurement section.

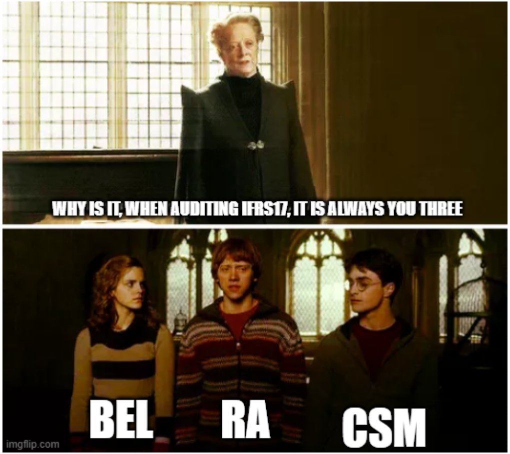
Contractual Service Margin (CSM) and Loss Component (LC) II

CSM and LC at Inception



Quick Break 2.0

In insurance business, we will be friend with these 3 items



Measurement Model

As we may have known, there are a lot of type of insurance. For example, we know there is a traditional life insurance, which covers risk of death, there is a marine hull insurance, which cover the risk of damage to vessel's hull and machinery, travel insurance and many more. We also learned so far that there are some complex items to calculate the LfRC. In this subsection, we are going to learn only the tip of the iceberg, eligibility to use the measurement model, on the other 2 measurement model aside of the General Measurement Model (GMM) and what type of insurance that must or eligible to use such measurement model.
There are 3 measurement model, i.e.,

1. General Measurement Model (GMM): insurance contract without direct participation features
2. Premium Allocation Approach (PAA): short-coverage insurance
3. Variable Fee Approach (VFA): insurance contract with direct participation features

VFA

VFA — Direct participating contracts

Premium Allocation Approach (PAA) I

An insurer may use Premium Allocation Approach (PAA) if, and only if, at the inception of the group:

1. the entity reasonably expects that such simplification would produce a measurement of the LfRC for the group that would **not differ materially** from the one that would be produced applying the **GMM**; or
2. the **coverage** period of each contract in the group (including insurance contract services arising from all premiums within the contract boundary determined at that date applying paragraph 34) is **one year or less**.

Source: IFRS 17, para. 53

The component of PAA at initial recognition is:

$$\text{PAA} = \text{Premium Received} - \text{Acquisition Cost} \tag{5}$$

Source: IFRS 17, para. 55

Which is way simpler than GMM model.

Transition Approach

It is not news that IFRS17 was published recently, and the old standard is IFRS4, so this section might not be relevant as time goes by since eventually all the insurance contract will used one particular transisition approach i.e., Full Retrospective Approach (FRA). But this transition approach can be useful later if one is doing Merger & Acquisiton (M&A). To put it simply, there are 3 transition approach, i.e.,:

1. Full Retrospective Approach (FRA): insurance contract is measured under IFRS 17 since the contract issued.
2. Fair Value Approach (FVA): insurance contract is measured under IFRS 17 at transition date.
3. Modified Retrospective Approach (MRA): insurance contract is measured under IFRS 17 before transition date.

Remeasurement

IFRS 17 actuarial audit

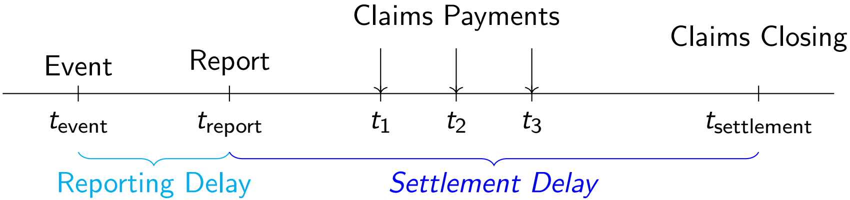
Life of a claim

The direct impact of issuing an insurance contract is to pay claim. But the claim itself has a lifecylce from accident to happened to insurance pay the claim. insert exam 5 topic on claim lifecylce

Liability for Incurred Claim (LfIC)

It is clear from earlier definition that there should not be a Liability for Incurred Claim (LfIC) at initial recognition since it implies that there is a claim directly after issuing the contract. LfIC only recorded after initial recognition or remeasurement.

IBNR



This event is what we call Incurred But Not (yet) Reported (IBNR). All IBNR claims will be the insurer's liability that will be reported in the financial statements. So the insurer needs to reserve the **right amount** of claim so that the reserve won't be too small (i.e., the insurer will not be able to pay claims) nor too large (i.e., the insurer's liability will be overstated).
Include reference to exam 5 material.

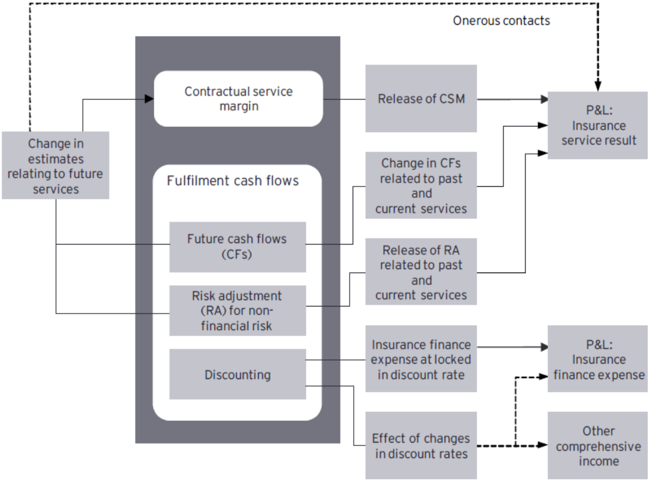
Financial Disclosure and Statement Mapping

IFRS 17 actuarial audit

Presentation

We have learned how to calculate all the need and the next step is to record everything inside the balance sheet. The next logical question is that, in the case where we do remeasurement, where should we record the changes ?

Graph from EY IFRS17 Playbook



Closing Remarks

References

- International Accounting Standards Board (IASB). *IFRS 17 Insurance Contracts*. IFRS Foundation, issued 2017 (effective 1 January 2023).
- EY. *Applying IFRS 17: A closer look at the new Insurance Contracts Standard* ("EY IFRS 17 playbook" / guide).